What is GIT in Hindi/Urdu | What is github | Distributed Version Control System

What is git

git 1 software ha jo k open source ha . Git ek mashhoor aur open-source Distributed Version Control System (DVCS) hai.

- git local py kam krty hn .git ak services ha .
- Release Date: Git ko April 2005 mein release kiya gaya tha.
- Git ko Linus Torvalds ne banaya tha (yeh wahi shakhs hain jinhone Linux Kernel bhi banaya tha)
- 2002 mein Linux team ne code ko manage karne ke liye BitKeeper naam ka ek proprietary (paid/commercial) version control tool use karna shuru kiya.
- 2005 mein BitKeeper company aur Linux open-source community ke beech ek conflict ho gaya, jiske natije mein BitKeeper ne Linux team se apna free usage access wapas le liya.
- Linus Torvalds ko Linux Kernel ke hazaaron developers ke code ko manage karne ke liye ek naye tool ki zaroorat thi. Us waqt ke baaki tools (jaise CVS ya Subversion) bohot slow the aur bade projects ka bojh nahi utha sakte the.
- Linus Torvalds ne sirf kuch hi hafton (weeks) ke andar ek naya, super-fast, aur distributed system khud design kar ke likh diya—jise unhone Git ka naam diya.
- **Extreme Speed**: System itna fast ho ki hazaron files par ek second mein kaam kar sake.
- **Data Integrity**: Code mein koi bhi change ho toh usme kisi kism ki corruption na aaye aur security maintain rahe.

What is github?

GitHub ek online platform aur **cloud-based web service** hai jo Git repositories ko internet par host karne, manage karne, aur doosron ke sath share karne ke liye istemal krty hai.

- github **microsoft** walo ki service ha .
- github storage provider ha jo hmry cord ko stor krta ha or hm zarort k wakt wha sy cord ko ly skty hn hithub py log apna pan cord public rakhty hn koi b istmal kr skta ha .

git na ho to kya ho ga ?

Bina Git ke: Jab aap code mein koi naya feature add karte hain, toh aksar file ko save karte hain jaise script_v1.py, script_final.py, script_final_updated.py. Agar kal ko code kharab ho jaye, toh purani version dhoondna bohot mushkil hota hai.

git k any sy kya hoa ?

Git ke Sath: Git aapke project ke liye ek Time Machine ki tarah kaam karta hai. Aap jab bhi code mein koi badlaav karte hain, Git uska ek snapshot (save point) le leta hai jise Commit kehte hain. Agar 10 din baad aapka code kisi galti ki wajah se crash ho jaye, toh aap ek single command se purane working code par wapas ja sakte hain.

Version Control

is ka matlab hota ha k mn ny pehly apny cord mn data lkikha tha (x,y,z) ya myra version 1 ho gya ha or (a,b,c) bnai ha is ka matlab ha jo mn ny dosra cord mn changing ki ha to os ka virsion 2 bn gya ha yani cord mn hr new hony wali changing ka 1 new version bna jata ha is ko version control system khty hn .

Distributed System

Distributed System (Taqseem shuda nizam) ek aisa system hota hai jisme computers ya devices alag-alag jagah par hote hain lekin aapas mein network ke zariye jude hote hain aur ek hi maqsad ke liye mil kar kaam karte hain.

- agr sab devices ki repo delete b ho jay to on ka apna apna data on ki **.git*** ki local repo mn mojod hota ha .

Branching & Merging

- Jab hm kisi project par kaam kar rahe hote hain, toh aapka main code ek safe jagah par hota hai (jise default taur par **main ya master** branch kaha jata hai).
- Jab aap apni nayi branch par kaam mukammal kar lete hain aur test kar lete hain ke code bilkul theek chal raha hai, toh aap chahte hain ke yeh naya kaam ab asal project (main branch) ka hissa ban jaye
- Apni nayi branch ke changes ko wapas main branch ke sath jodne (combine karne) ke process ko**Merge kehte** hain

Team Collaboration

Hazaron log ek saath ek hi project par kaam kar sakte hain bina ek dusre ke code ko overwrite kiye.

git & github

- **Git**: Yeh ek tool/software hai jo aapke computer par locally chalta hai aur version track karta hai.
- **GitHub**: Yeh ek cloud website/service hai jahan aap apna Git repository upload (push) karte hain taaki doosron ke sath share kar sakein ya backup rakh sakein.

Nayi repository shuru karne ke liye kya krty hn ?

```
git init
```

Files ko save karne ke liye select karne ke liye

```
git add
```

Badlaav ko history mein snapshot ki tarah save karne ke liye

```
git commit -m "message"
```

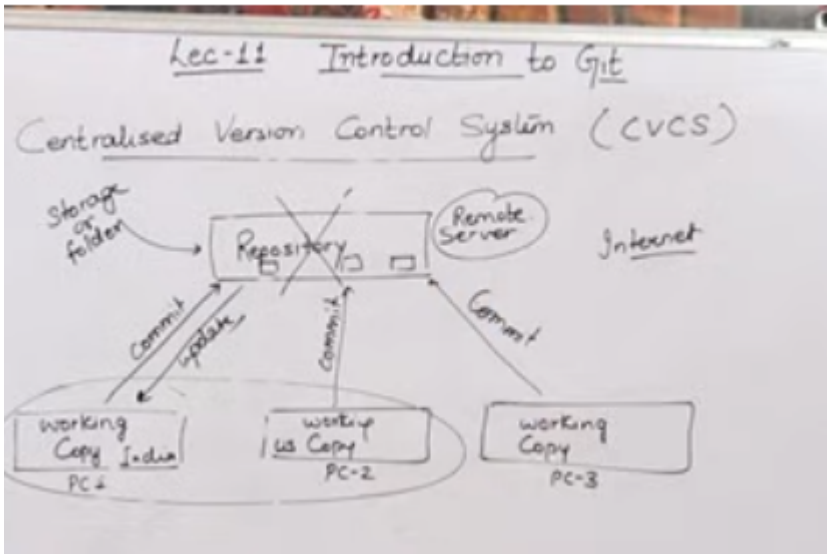
Dekhne ke liye ki kaun si files change hui hain

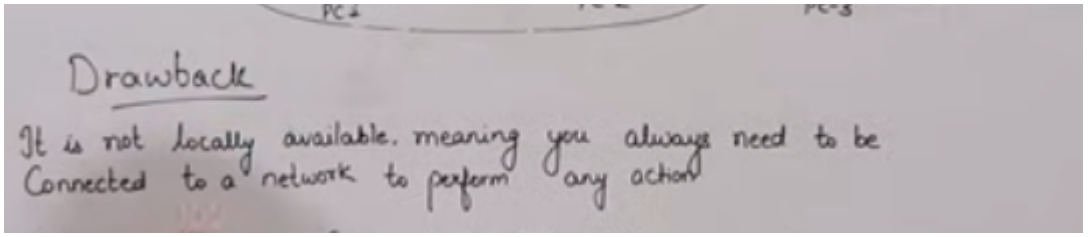
```
git status
```

Central Repository

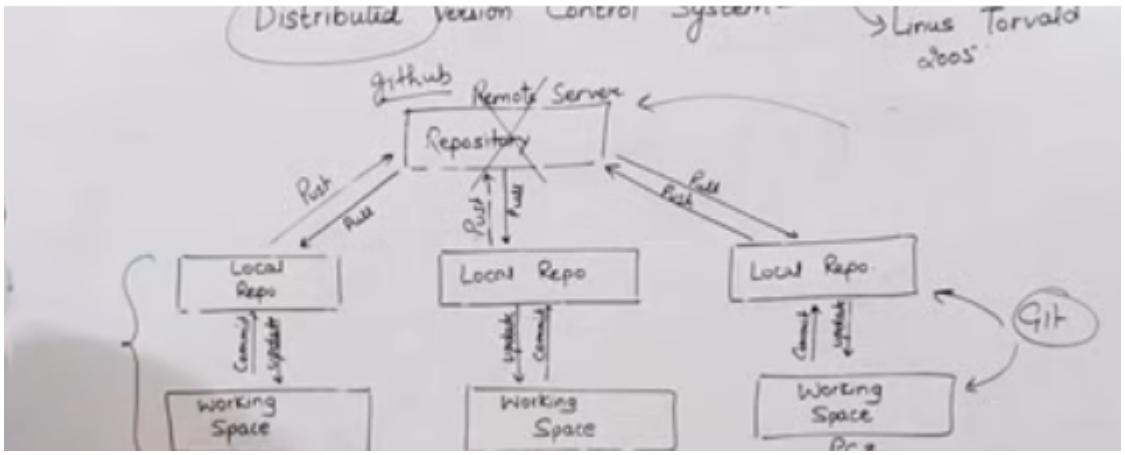
Central Repository ek aisi main (central) jagah ya server hota hai jahan kisi project ka saara code, uski poori history, aur saari branches secure tareeqe se save hoti hain.

- apny cord ko github py save krny ko **git comit** khty hn .
- sab sy barda faida ya hota tha k mn ny agr dykhna ha k baki k 3 log kya kam kr rhy hn to mn on ky dono k gitcomit ko dykho ga k oono ny kya kam kiya ha is trha mn on ka cord b dykh sko ga . or is sy pata chlta ha k mn kya kam kr rha ho or dosry 2 log kya kam kr rhy hn.hm same hi kam to nhi kr rhy .
- is ka matlab ha myri centeral repo sy dykh skta ha .
- is ka hm ko 1 noksan kya hota ha k mn ny jo b repo mn cord ko change kr k save kiyta tha os ko update krta rha tha lakin os ka myry pas backup nhi ha agr myri repo hi krab ho jatoi ha is ka mujy bht barda noksan ho ga k myra sara cord zaya ho jay ga ya myra server hi down ho jay jis pr myra cord parda tha .
- mujy aws ko chalany k liyy internet ki b zarort hoti the bina internet k to hm kam b nhi kr skty thy internet bina hmy update lyny mn or cord ko upload krny mn bht mushkl ka samna krna prdta tha .





- hmary pas 1 local repo hoti ha jo hmara sara data apny 1 folder mn save krrti ha os folder ka name ha **.git** is k andr hmara sara data yani hmari configration sara data or history mojod hoti ha .
- agr hmari repo na b ho to hmary system mn hmari apni local repo ka 1 folder hota ha jo hmara data ko save rakhta ha .



- client jo ha wo apny local py push krta ha pr apny public is k bad apny hithub py push krta ha dono jga pr save ho jata ha data . -**CVCS** is mn 2 repo aps mn conflict ho skti hn .or ya bina internet k is py kam nhi kr skty . hmy hr time centerl server sy conect hona prdta ha .

- Agr **CVCS** server down ho jata ha to devolper kam nhi kr skty on ka data loss ho jayy ga .
- **DVCS** is mn hmara cord hmary local repo mn prda hota ha or hm jb dil kry tb hi os ko ota kr apny hithub py upload kr skty hn . or is ko internet ki zarort nhi hoti . is ko apni hi hard disk ky sath conect ho kr kam krna hota ha is liyy fast hota ha .
- agr **DVCS** server down ho jata ha to apni local repo sy dobara data ko ly ly ga or or data loss nhi ho ga.

فاصلے ایسے بھی ہوں گے کبھی سوچا نہ تھا

سامنے بیٹھا تھا میرے اور وہ میرا نہ تھا

Lecture No:2 Git-Three Stages -Working Directory,Staging area and Local Repository | What is Branch in git

local version control system

- sab sy pehly hmary pas CVCS tha is mn ya hota tha k hmara kisi or jga pr 1 server parda hota tha is mn hm apna data ko push kr dyty thy lakin masla ya tha k agr wo server drab ho jata ha to hmara sara data loss ho jay ga .

- is masly ko hal krny k liyy hm ny DVCS byana ha is mn ya hota ha k hamra jo data kisi or jga kr server mn mojo ha is k ilava hmary server mn b hojod hota ha or hr 1 server k pas apna local data ki first copy mojod hoti ha agr jha pr hmara data parda ha wha sy delet ho b jay to hmary local server py data mojod hota ha hm vha sy os ki coppy ly k apna data ho dobara sy istmal kr skty hn jis ka name (git, mercurial) mercurial ya abi itna zyada istmal nhi hota ha .

- hmary pas ak or service ha jis ka name gitlab ha ya central storage ha jha pr mn apny data ko store kr k rakh skta ho.

- farz kro myry pas 2 machine hn 1 mobai mn ha or 2sri north virganiya mn yha pr command ko run kro ga ginstall kro ga . git free ha . hm 1 folder bnay gy os ko name dy gy myrepo is k andr mn 1 command ko run kro ga `git init` yani mn git ko start krna chahta ho. jb ya comand chly gi is ky andr ak hidan folder bn jay ga **.git** k name sy or jo mn ny folder bnaya tha wo convert ho jay ga **git reposetry & local reposrtory** k andr . os k bad yha pr 3 chezy bn jay gi . **1st** working diractory & workspace bn jay gi **2nd** staging aria **3rd** local repo

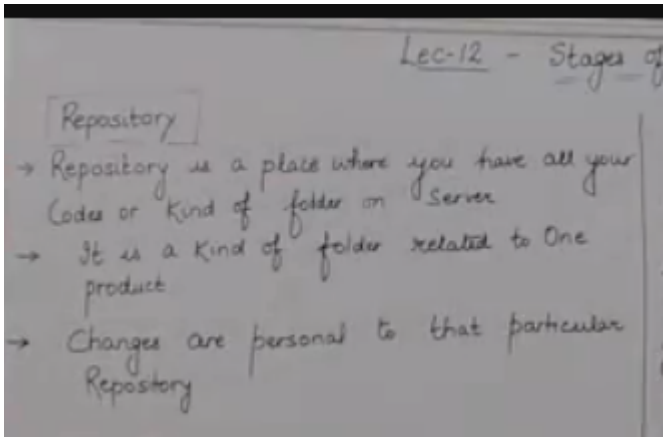
- mn jo b cord likho ga wo ho ga hmara working diractory

- jb mujy pata chjal jay k mn ny jo cord likha ha k mn ny cord pora likh liya ha to mn os ko add krna chahty hn to is ko staging aria khty hn.

- satging aria mn krny k bad mn is ko comit kro ga myr pas agr 3 file mojod hn mn on mn sy jis jis ko select kro ga on ko mn git comit kr lo ga . comit krny ka matlab ya hota ha k mn ny apna jo cord ha os ko ny save kr liya ha yani jb b mujy zarort ho gi mn is ko dobara sy istmal kr lo ga . **snapshort** mn khty hn . jo mn ny cord ko save kiya ha os ko ak ID mila jay gi jis ko hm **comit ID** khty hn .

- comit id unik hote ha hm ny jb b hmari jisi porany cord py jana ha hm os ki comit id ko lga kr dobara sy cord ko dobara sy dykh skty hn. is mn hmary pas 40 alfa bats hoty hn. mn ny apny comit id ko tage b dy

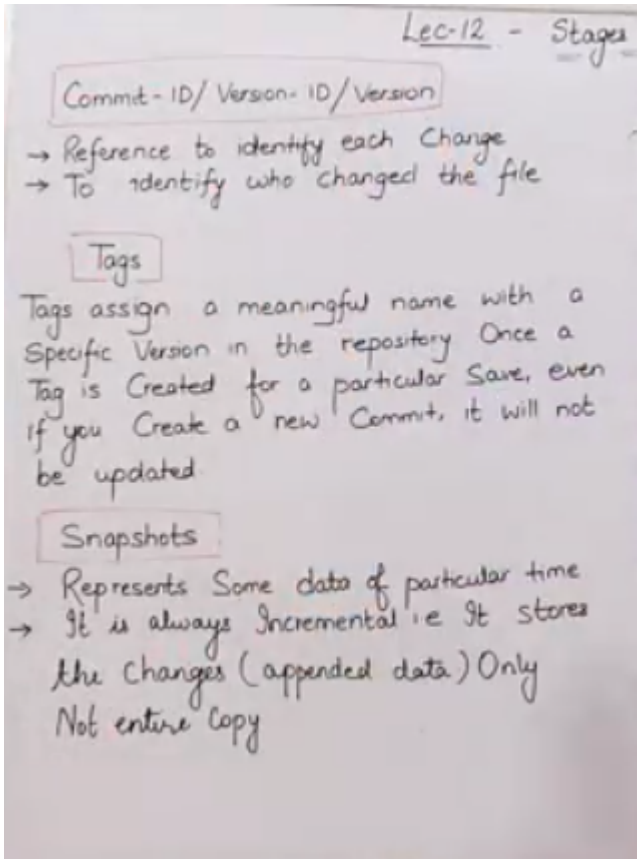
skta ho jis sy mujy asani ho gi apni comit id ko dondny mn .

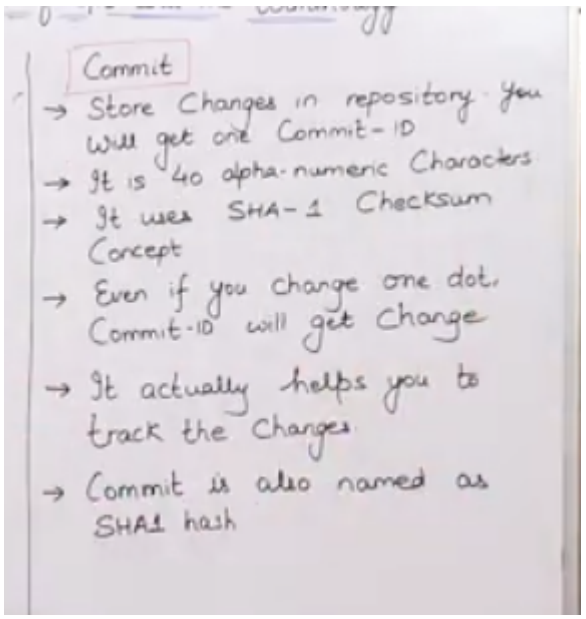
- github py public or private reposatery hoti hn.
- isi trha hmary pas gitlab b ha jo k open sorce ha public or private dono trha sy hi.
- jb hmara local repo mn kam ho jay ga hm is cord ko github mn push kr dy gy .
- git sy github py jany k liyy hmy internet ki zarot prdti ha .
- ab hmara cord hmari local mn b ha or hmari github py b ha . jb hm cord ko local repo mn ly jaty hn to wo hmari master branch mn jata ha jo k bydfalt master branch hoti ha
- myri ab dori company ka banda jo ha wo singapor mn ha is ny ec2 instance bai hovi ha.
- ya ab kya kam kry ga k github y pul kry ga cord ko to first time sara data ko pull kr ly ga .
- same is ny b yhi kam kiya cord ko clone kiya add kiya pr comit kr diya . is k bad is ny cord ko dobara sy kam pora kr k githu py push kr diya . push krny sy hmari file mn jo data add hoa ha os mn changing ho gai ha or hmary pas update data mojod ho ga . pehly wala banda jb is cord ko dobara sy pull kry ga to hmary pas data update ho jay ga . same isi trha sy ya dono bndy is cor mn update krty rhy gy or dono ko hi data ka pta chlta rhy ga k cord mn kya changings hovi hn or is sy kam b asan ho ga o data contribution ho jayy gi .
- jo b hm hr bar changings kry gy to os mn hmary pas hr kisi ki comit ID defirat hoti ha jis sy hm hr comit ko dykh skty hn kya changing hovi ha .
- hamry pas version controll system sy pehly kya tha k jb hm apny data ka snapshot lyty thy to hmari agr pehli file 1GB ka ha to hmari dosri changing wali file ko dobara sy snapshot lyny sy o ka data 1.5GB ho gya is ka matlab ya hoa k hr bar hmy alag alag sy data mn stor krna prdta tha or hmari storage ka bht zyada istmal hota tha or hmy is mn bht mushkl hoti the .
- 

- is mn mayr pas jo rposetery ha ya loacal b ho skti ha or central b ho skti ha agr hamry servr py ha to ya local ha agr github py ha to public ya private b ho skti ha .
- har repo mn alag alag cord ho ga .
- hm jo b chnging kry gy hmari reposatery mn hi ho gy .
- hmar pas 1 server b hota ha agr github py ho ga is ka matlab ha k microsoft wala sever ho ga agr hmari machine ka server ho ga to is ka matlab ho ga k hmair VM ka server ho ga . is k andr hmara meta data

hota ha jo k hami total hony wali jitni changings hn on ko b save krta ha . yani do hm ny comit dmata kiya ha os ki comit ID kya ha os ka tage kya ha yani tagces ko b save krta ha.

- is mn hmay pas working diractory hoti ha jha pr hm apny cord ko likh skty hn modify b kr skty hn .
- hm ak time mn 1 hi kam kr skty hn yani 1 branch mn 1 time pr changing kr skty hn jo .
- CVCS mn ya hota tha k workspace mn kam krty thy or direct comit kr dyty thy os mn koi staging aria nhi hota .
- lakin hamry case mn ya hota ha working aria hota ha . git mn ya hota ha k ya hmari os workspace wali diractory ko nhi dykhy ga wo sirf hmari staging aria wali files ko hi dykhy ga . hm local sy apni github py b push kr skty hn . jo k hmari central reposatery ha .



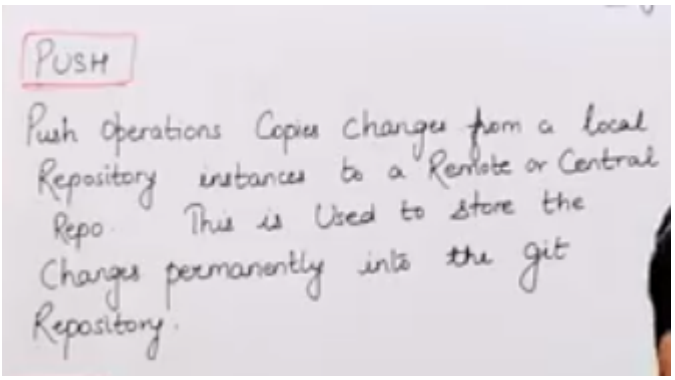


- jb hm git comit krty hn to wo hmari alfanomaric fom mn hota ha . wo a.b.c.d or 1.2.3.4 or @.! form mn hota ha .

SHA-1 (Secure Hash Algorithm 1) Kya Hai Git Mein?

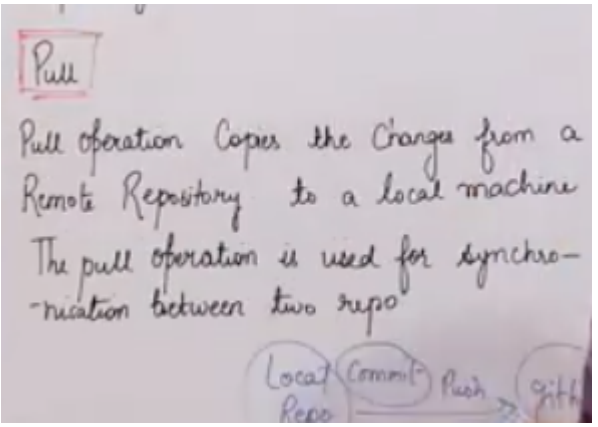
Agar aap apne code mein ek chota sa comma (,) ya space bhi add ya delete karenge, toh poora SHA-1 code badal jayega. Iska fayda yeh hai ke koi bhi chori-chhupe purana code ya history change nahi kar sakta.

- is mn hmary pas 40 cheractors ka ek unique cord hota ha alphaunmaric cord banata ha .
- `d670460b4b4aece5915caf5c68d12f560a9fe3e4`
- **comit ID** sy hm ak ak changing ko track kr skty hn .is sy hm ya b adentifi kr skty hn k ya kis user ki ha or kis ny is ko change kiya ha .
- **Tangs** is liyy diya jata ha k hmary pas jo b comt ID ho gi os ki pechan k liyy hm mekhsos tage diya jata ha taky hm apny comit ki pechan kr sky .
- **Snapshots** is ya hota ha k hm ny kisi mehsos time pr comit kiya tha snapshot liya tha os time ka data dikhayy ga or os time ki change wala data jo save kiya ha os ko hi dikhayy ga .ya b incrimantel hota ha . comit mn hm nai nai chozo ko add krty hn apand krty hn . azafa krty hn . ya on file ka snapshots nhi ly ga jis mn hm ny koi changing nhi ki jis mn koi b changing hoi ha os ko save kr ly ga yani apand kr ly ga .



push

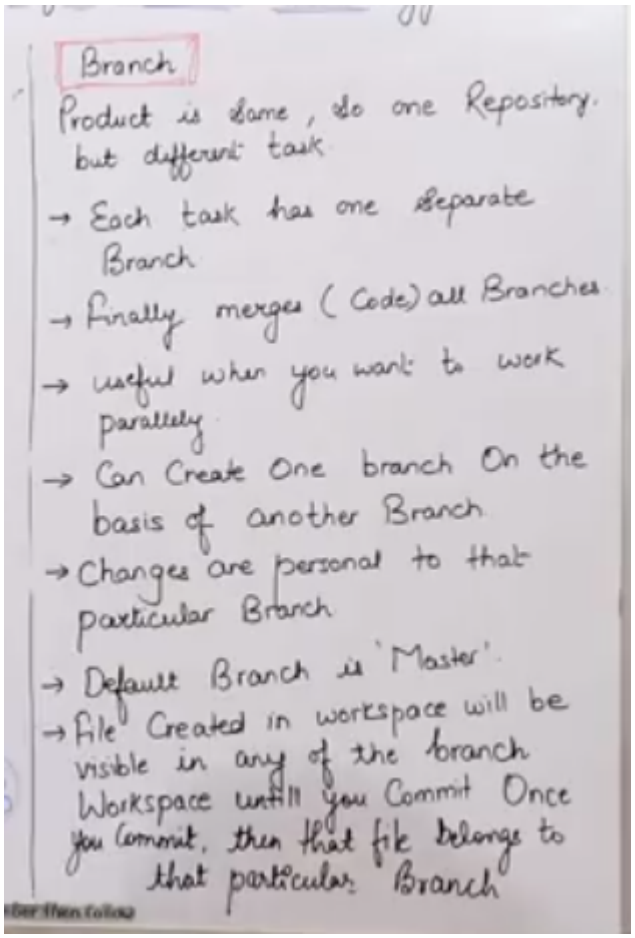
- hm ny jo b data ko local sy github pr byjy gy os ko hm push khty hn. yani hmary data ko copy krta ha or os data ko hmary server ya github py save kr dyta ha .



pull

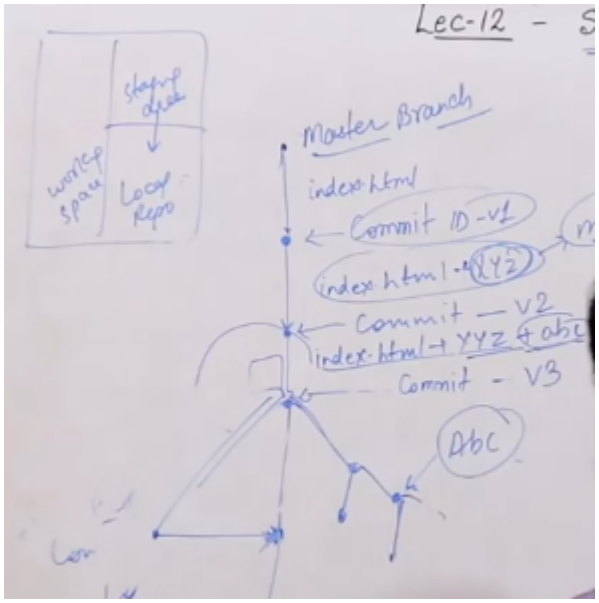
pull ka matlab hota ha (central reposatery) github , gitlab sy cord ko hmari local machin mn ly k ana is ko pull khty hn . yani myri jo b kisi ny new changings ki hn wo wala data agr hm pull ly gy to data hmary pas same to same hmari local machine yani instance mn a jay ga .

- pull krny sy hmary pas sincronise data milta ha yani jo b changing dosri sid waly bndy ny ki ha wo hmari trf b nazr a jay gi . hm ny jo changing kr k push kiya ha data wo b os ko nazr a jay ga .



- **Master Branch** mn jo b comit kro ga wo myri master branch mn a jay ga . mn ny 1 comit kiya 1 comit ID a gai ha . mn ny indek.html file ko likha . is k bad mn ny 1 or naya cord likha os mn changing ki . ,mn ny new file bnai os ki changing ko comit kiya to myra dosra version bn gya jo pehly file bnai the wo pehla version bna tha . mn ya jitni bar b comit krta ho wo sara myri master branch mn ho ga . mn per sy 1 or new file bnao ga ya or os ko comit kry ya b hmari master branch mn kam ho ga .
- 1 time aysa ayy ga k hmara client hm sy khy ga k to version 2 ha wo acha ha os ka cord or os mn kiya hoa kam acha ha os agr tum ny koi or changing krni ha to mn tumhy is k liyy 1 or chlid branch bna k dyta ho tum os mn jo b kam krna ha os ko kro myra is cord mn koi changing na krna mn koi risk nhi lyna chahta .
- agr os bndy kam acha hoa to mn ny jo file a.b.c jis mn osy changing nhi krny di the os ko new bnai hovi file k sath replace kr do ga . ab mn chlid branch k kam ko master branch k sath marg kr do ga .
- agr mujy os bndy ka kam acha nhi laga ho ga to mujy koi fark nhi ho ga . ku k myra sara kam runing instance mn chal rha ha or jo changing hovi ha wo to child branch mn hovi ha .
- jb hm child branch bnaty hn .
- agr hm ny bht sari child branch bnai hn or sab branch mn alag alag log kam kr rhy hn agr hmm ny master branch mn koi kam add krna ha yani marg krna ha to hm jo jo kam acha ha hm os kam ko add kr ly yg jis sy hmari application chaly gi . is trha hm sab ki branch ko add kr k hm sab ka kam akta kry gy ap application bna dy gy .
- agr hm ny ak branch bnai ha or jb hm os mn koi data add kry gy to os ka data hmary pas save ho jay ga. agr hm ny koi or branch bnai ha os ko jb hm comit kry gy to hmary pas master branch ka data copy

nhi ho ga blky os branch ka data hi copy ho ga jis branch sy hm ny data liya ha .



types of repositories

Local Repository

Local repository woh hoti hai jo aapke apne computer ya laptop mein hoti hai.

Remote Repository

Remote repository kisi remote server ya online platform par hoti hai, jahan project ka code store aur share kiya jata hai.

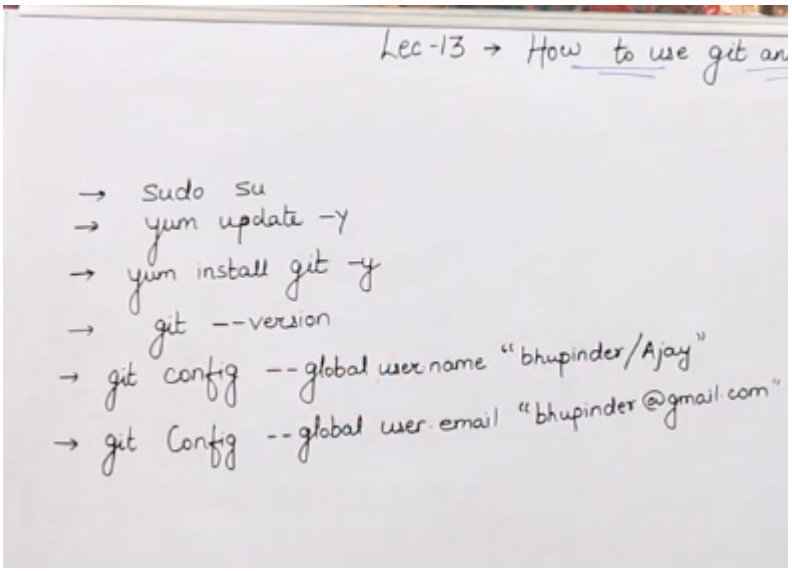
> بچھڑ گئے مگر یاد آنا چاہیے تھا نام نہ سہی، چہرہ یاد آنا چاہیے تھا۔ حوصلہ بہت تھا، میری آنکھیں چھلک آئیں، میری محفل میں تیرا میرے بعد آنا چاہیے تھا۔

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lecture No: How to install git on aws linux machine | How to Create github account | What is git?

- sab sy pehly hm apny aws account mn ec2 instance bnayy gy os k bad hm apni vm ky port 22 ssh k liyy open kry gy or os k bad hm http port 80 ko b open kry gy .
- hm apna ssh ly gy puty sy ya pr hm mobaxterm py apni private key sy ly gy . os k bad hm apny system ko `sudo -i` ki command sy root user bnayy gy .
- git ko install krny sy pehly hn apny systemm ko `yum update -y` sy update kr ly gy .
- ab hm git ko install krny ki command ko run kry gy `yum install git -y` jysy k hmy pata ha k hmary linux ko bnany wala bnda linux torwald ha os ny hi git ko bnaya tha is liyy hmary pas system mn git ki files mojod hoti han lakin hmy is ko install krna prdta ha .
- git `cli` hota ha .
- agr koi hm sy pochy k ap ka git ka konsa version ha to hm ko pta hona chahyy k hm konsa git ka version use kr rhy hn git ka version dykhny k liyyy hm ya command likhy gy `git --version` .
- agr mn ny dykhna ha k myry pas git install ha k nhi to hm `which git`

- ab hm git ko 1 @gmail dy gy or user name dy gy is sy pata chly ga k is bndy ny ya comit kiya ha or ya os ki email ID ha . agr kisi ko koi masla ha cord mn to wo mujy myri email py send kr skta ha . ya sara kam hmari centeral repositery mn ho rha hota ha jo k hr koi dykh skta ha . agr hm ny apna user name ko add krna ha to hm `git config --global user name "meharbadshah"` ab agr hm ny apni email ko add krna ha to hm is ko hm `git config --global user email "meharbadshah686@gmail.com"` aysy add kry gy .
- myry pas 2 machine hn or mn ny on pr git ko b installl kr liya ha .
- mn ab `www.github.com` py jao ga or mn apna 1 account creat kro ga . git hmari centeral repository ha



- mn ny ab git ko install kr liya ha ab mn ny dykhna ha k myra git install kha pr hoa ha to mn `which git` ki command ko run kro ga

```
[root@ip-172-31-46-110 ec2-user]# which git
/bin/git
[root@ip-172-31-46-110 ec2-user]#
```

- agr mn ny dykhna ha k myra git ka konsa version ha `git --version` ya command ko run kry gy

```
[root@ip-172-31-46-110 ec2-user]# git --version
git version 2.52.0
[root@ip-172-31-46-110 ec2-user]#
```

- ab mn apna user name ko add krny laga ho is k liyy ya command o rum kro ga `git config --global user.name "meharbadshah"`
- ab mn ny agr email ko add krna ha os k liyyy `git config --global user.email "meharbadshah686@gmail.com"` ki command ko run kro ga .
- agr mn ny dykha ha k myra user name or email add ho gya ha k nhi `git config --list` to mn ya command ko run kro ga .

```
[root@ip-172-31-46-110 ec2-user]# git config --list
user.name=meharbadshah
user.email=meharbadshah686@gmail.com
[root@ip-172-31-46-110 ec2-user]#
```

- ab mn apni dosri machine mn jao ga os ka mn region ko change kr k os pr b sara same kam kro ga .
- ab mn user name ko add krny laga ho `git config --global user.name "meharumar"`
- is k bad mn ny apni email ko add krna ha . `git config --global user.email "meharbadshah5218@gmail.com"` is trha mn add kro ga .

- mn ny ab list krna ha to mn `git config --list` ko run kro ga

```
[root@ip-172-31-41-181 ec2-user]# git  config --list
user.name=meharumar
user.email=meharbadshah5218@gmail.com
[root@ip-172-31-41-181 ec2-user]#
```

---

lectre NO : 4 - How to add,commit in git | How to push from git to github | How to pull code from github

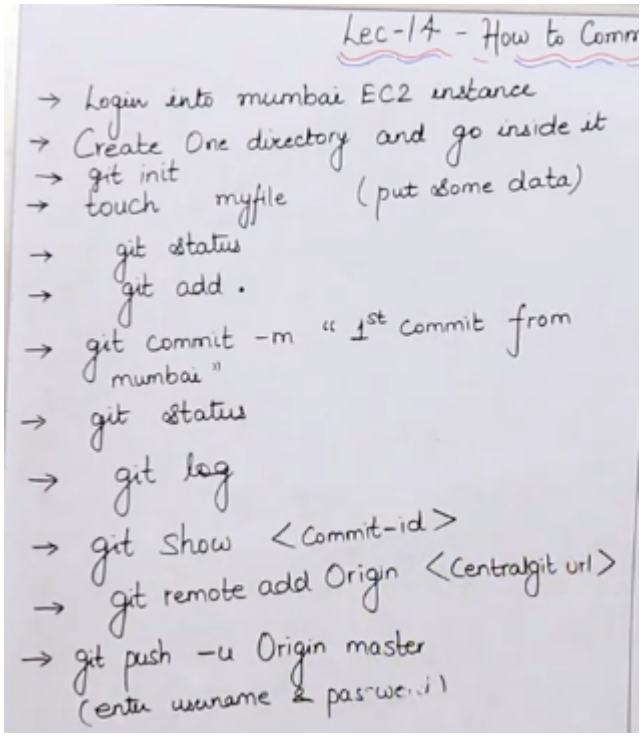
DVCS kya ha ?

DVCS (Distributed Version Control System) Is system mein har developer ke computer par repository ki complete copy hoti hai, jis mein project ka code aur uski history shamil hoti hai. Developer local level par bhi commits kar sakta hai.

CVCS kya ha ?

CVCS (Centralized Version Control System) Is system mein project ki main repository ek central server par hoti hai. Developers us server se code access aur changes share karte hain.

- sab sy pejly hm ec2 ko chlayy gy os k bad hm os k andr 1 folder bnay gy jis k andr hm `git init` ki command ko run kry gy . jb mn git init ki command ko run kro ga to wo folder git repo mn convert ho jay gi
- mn is k bad touch sy 1 file bnao ga `my file` os k andr cord ko likho ga .
- os k bad mn status chek kro ga `git status`



is k bad agr hm ny git mn add krna ha to `git add .` ki command ko run kry gy .

- commit kry k liyy `git commit -m "1st commit for lahore"` ab hamry pas foler mn koch b nhi bachy ga ku k hm ny sab koch ko comit kr diya ha .
- ab hm agr pr sy dobara sy `git status` kr k dykhy to hmy koch b nhi mily ga .
- is k bad `git log` is sy hmy pata chlta ha k kis ny cord ko commit kiya ha kb or kis time comit kiya ha .
- agr hm ny dykhna ha kisi user ny commit kiya ha to hm `git show <comit-id>` ki comand ko run kry gy . is sy hmy sari details mil jay gi.
- ab mujy ya sara kam githun mn phonchana ha to mn `git remote add oragin <central git url>`
- is k bad hm ny `git push -u origan master` ki command ko run kr dy apna sara cord **github** py byjna ha . is mn hmary pas **-u** user k liyy hota ha . or origan mn hmy master brach mn push krna ha .
- is k bad ya hm sy username or password pochy ga . pr wo hm sy github account ka password pochy ga . or hm conect ho jay gy .
- mn jb apni file ko git status ki command ko run kro ga to mujy koch aysa hi nazr ayy ga

```
[root@ip-172-31-41-181 mehar]# cat >badshah
dill a nada tujhy hoa kya ha [root@ip-172-31-41-181 mehar]# cat badshah
dill a nada tujhy hoa kya ha [root@ip-172-31-41-181 mehar]# git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        badshah

nothing added to commit but untracked files present (use "git add" to track)
```

  is ka matlab ha mn ny abi koch comit nhi kiya jo is mn red nazr a rha ha ya koch new likha ha os ko abi add kiya ja skta ha abi add hony wala ha .

- ab mn is ko add kro ga `git add .` ki command ko run kro ga .
  - mn ny jb git add kr liya to is k bad mn ny `git status` ki command ko run kiya to mujy ab sab nazr a rha ha .

```
[root@ip-172-31-41-181 mehar]# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   badshah

[root@ip-172-31-41-181 mehar]#
```

  - ab k bad mn ny jb commit kiya is command sy `git commit -m "my first commit for badshah"`
  - is sy pata chly ga

- ab mn `git log` ki command sy apni git ki ID ko dykho ga .

```
[root@ip-172-31-41-181 mehar]# git log
commit 3dd5a7889b64f4ea5dd132f534d31e3c11f9ba33 (HEAD -> master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Sun Oct 18 12:01:15 2026 +0000

    my first commit for badshah
[root@ip-172-31-41-181 mehar]#
```

is pic mn sab nazr a rha ha mn ny kb commit kiya tha myri gmail b nazr a rhi ha agr koi mujh sy contect krna chahta ha to myri email py rabta b kr skta ha . mn ny jo massege jo diya tha wo b nazr a rha ha my first commit badshah is trha ka likha nazr a rha ha .

- agr mn ny dykha ha k mn ny jo commit kiya thsa os k ander kya likha ha to mn ya command ko run kro ga `git show 3dd5a7889b64f4ea5dd132f534d31e3c11f9ba33` git show os k bad mn apni comit id ko add kro ga

```
[root@ip-172-31-41-181 mehar]# git show 3dd5a7889b64f4ea5dd132f534d31e3c11f9ba33
commit 3dd5a7889b64f4ea5dd132f534d31e3c11f9ba33 (HEAD -> master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Sun Oct 18 12:01:15 2026 +0000

    my first commit for badshah

diff --git a/badshah b/badshah
new file mode 100644
index 0000000..8a1eaaf
--- /dev/null
+++ b/badshah
@@ -0,0 +1 @@
+dill a nada tujhy hoa kya ha
\ No newline at end of file
[root@ip-172-31-41-181 mehar]#
```

ya mn ny is mn data likha tha

- ab mn apny is cord ko apni cernral reposatiery mn byjna chahyta ho to is k liyy mn apny github mn jay gy is k bad mn **create new repositery** py click kr k os mn jao ga. or apni repo ka name rakho ga **central repositery** apny destination mn `koch b likh skty hn` or kha pr hm publice or private dono mn sy 1 ko rakhy gy . agr mn github py public py koch rakho ga to mujy free mn use kr lo ga agr private krna ha to paid ho ga .

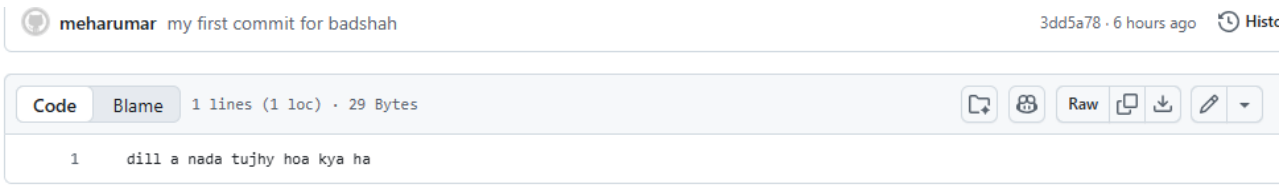
- ab is k bad mn ny apny apny ec2 mn jao ga is k bad `git remote add origin https://github.com/MeharBadshah5218/central-repositery.git`

```
[root@ip-172-31-41-181 mehar]# git remote add origin https://github.com/MeharBadshah5218/central-repositery.git
```

- mn ny `git push -u origin master` kiya is k bad mn ny apna user ko likha **meharumar** or bad mujh sy os ny token ka pocha mn ny pr token bnaya or mn ny os ka password ko add kiya or mn ny kam ko pora kr liya

```
[root@ip-172-31-41-181 mehar]# git push -u origin master
Username for 'https://github.com': meharumar
Password for 'https://meharumar@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 249 bytes | 249.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/MeharBadshah5218/central-repositery.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
[root@ip-172-31-41-181 mehar]#
```

- mn ny is k bad apny github ko refri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sh kiya or dykha k myra jo cord tha wo mujy wha pr likha hoa nazr a gya



is k bad mn ny apny cord mn or b koch add kiya ha k `cat >.....`

```
[root@ip-172-31-41-181 mehar]# git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   badshah

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        akhir

no changes added to commit (use "git add" and/or "git commit -a")
[root@ip-172-31-41-181 mehar]#
```

ab mn ny dobara sy `git status` is mn koch modifiey hoa ha is ka b dikha rha rha ha .

- ab mn `git add .` ya comand ko run kro ga

```
[root@ip-172-31-41-181 mehar]# git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   akhir
        modified:   badshah
```

mn ny git mn ad commit b krna ha `git commit -m "my second repo in pakistan"` to wo commit ho jay ga .

-

```
[root@ip-172-31-41-181 mehar]# git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
[root@ip-172-31-41-181 mehar]#
```

is k bad mn ny is mn dykha k likha ha yha pr ab koch b nhi ha git pr comit kr diya ha . hmary pas agr dykhna ha to hm new commit kya hoa ha `git log`

```
[root@ip-172-31-41-181 mehar]# git log
commit 3106e90e0d5bf2ef251a3576cab2e77b61d1f151 (HEAD -> master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Sun Oct 18 19:06:22 2026 +0000

    my second repo in pakistan

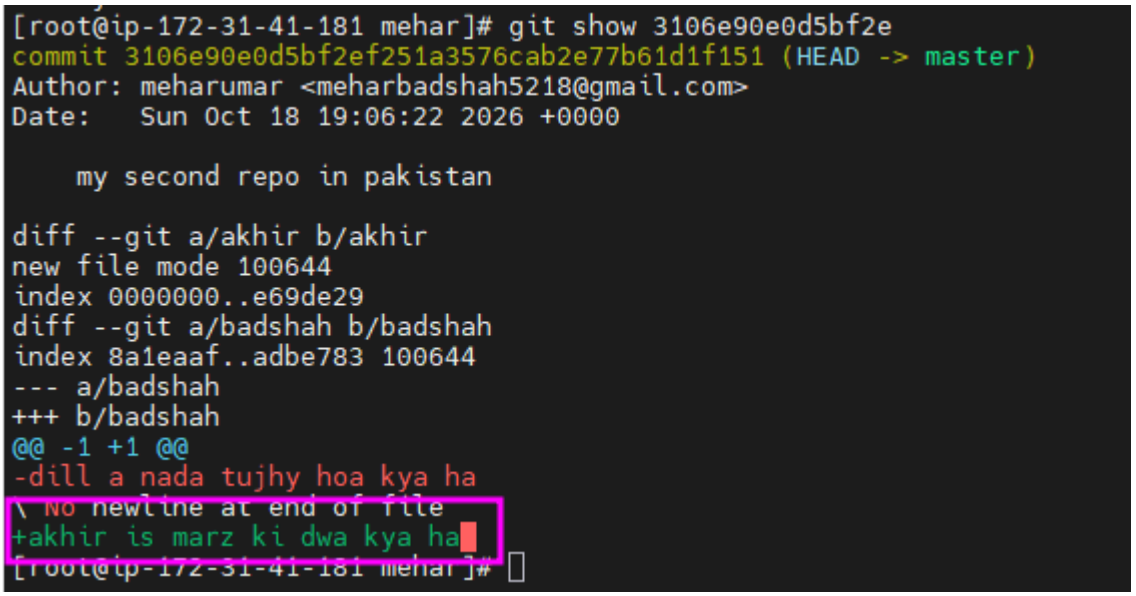
commit 3dd5a7889b64f4ea5dd132f534d31e3c11f9ba33 (origin/master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Sun Oct 18 12:01:15 2026 +0000

    my first commit for badshah
[root@ip-172-31-41-181 mehar]#
```

new commit ID

is mn jo opr ha wo new comit ID ha .

- is k bad mn ny pr sy `git show or commit ID` ko add kiya to mujy koch is trha sy jo mn ny new commit kiya tha nazr any laga



- my token (umar1234) `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`
- ab mn apni dosri machine mn jao ga or os ka ssh lo ga .
- mn ny apni dosri machine mn ja kr ak folder bnaya `mkdir mehar` os k ander ja kr mn ny `git init` ki command ko run kiya . is k bad mn ny apni dosri ec2 ka pull lyna tha github sy . to mn ny `git remote set-url origin https://github.com/Me harBadshah5218/central-repositery.git`
- is k bad mn ny `git pull origin master` kiya to mn ny jo b pehly machine ,mn kam kiya tha wo sara myry pas is machine mn pull ho gya .

- mn ny `git log` kr k dykha to mujy sary jitny b commit hovy thy sab nazr any lag gay .

```
commit e36e2e57ffd4b2b13eab25fa8a33bbe31655c5d4 (HEAD -> master, origin/master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 06:19:18 2026 +0000

    Update project data

commit 4c9616031091f5978908e9f01cbdcd8bce609c07
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 06:17:54 2026 +0000

    Update project data

commit 7d4999def935bf0a02513e459413579e7df7dc90
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 06:14:48 2026 +0000

    Update project data

commit 84cb93ccffd0bfca5f0c7c6216bbe3c89f743efd
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 05:04:25 2026 +0000

    c7304c99beec65bc79c4c7be8a6468a8574d0816

commit c7304c99beec65bc79c4c7be8a6468a8574d0816
Merge: 09a647d fceccfd
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 04:58:55 2026 +0000

    Merge branch 'master' of https://github.com/MeharBadshah5217/central-repositery

commit 09a647d35347ecb623fcd6f188bebfb493267e14
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 04:36:18 2026 +0000

    my second repo commit

[root@ip-172-31-46-110 mehar]#
```

- ab mn b is k andr naya cord add kr do ga . is k liyy mn apny pas jo pull kiya ha os file mn or b data ko add kro ga . `cat >>badshah` kro ga or data add kr do ga .

- is k bad mn ny data ko add kr k `git add` kiya to new data b add ho gya

```
 (use "git restore --staged <file>..." to unstage)
       modified:   badshah
```

- is k bad mn ny git pr apna data ko b commit krna ha to mn `git commit -m "my last data in pakistan"` sy commit kr diya ha .

- is k bad mn ny `git log` kiya to mujy myra data commit kiya hoa nazr koch is trha ka aya

```
[root@ip-172-31-46-110 mehar]#  git log
commit b0ce9a05455a752eb4adeeda7d94c039cadd16e8 (HEAD -> master)
Author: meharbadshah <meharbadshah686@gmail.com>
Date:   Mon Oct 19 06:54:57 2026 +0000

    my last data in pakistan
```

- ab mn `git show commit ID` ko add kro ga to mujy myra data add hoa nazr ayy ga .

```
[root@ip-172-31-46-110 mehar]# git show b0ce9a05455a752eb4adeeda7d94c039cadd16e8
commit b0ce9a05455a752eb4adeeda7d94c039cadd16e8 (HEAD -> master)
Author: meharbadshah <meharbadshah686@gmail.com>
Date:   Mon Oct 19 06:54:57 2026 +0000

    my last data in pakistan

diff --git a/badshah b/badshah
index 149f317..49c40c9 100644
--- a/badshah
+++ b/badshah
@@ -1,2 +1,3 @@
 My new data
 my second data is addd
+jinhi malom hi nhi wafaa kya ha .
[root@ip-172-31-46-110 mehar]#
```

- is k bad mn ny apni centerl repo mn data ko push krna ha to mn command ko run kro ga `git push origin master` os k bad username or password do ga

- ec2 token `(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)` meharsahab

- mn ny apny dosry ec2 sy jo data add kiya tha os ko mn ny ab dykna ha k data aya ha k nhi to mn ya command ko run kro ha `git show commit ID`

```
+jinhi malom hi nhi wafaa kya ha .
[root@ip-172-31-46-110 mehar]#
```

```
1   My new data
2   my second data is addd
3   jinhi malom hi nhi wafaa kya ha .
```

is mn ab mya data b add ho gya ha jo mn ny add krna tha .

- mn ab apni pehli intance mn gya or pr sy mn ny pull krna ha k dosry bndy ny kya changing ki ha os ko dykhny k ;liyy

```
[root@ip-172-31-41-181 mehar]#  git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 318 bytes | 318.00 KiB/s, done.
From https://github.com/MeharBadshah5218/central-repositery
 * branch            master     -> FETCH_HEAD
   e36e2e5..b0ce9a0  master     -> origin/master
Updating e36e2e5..b0ce9a0
Fast-forward
 badshah | 1 +
 1 file changed, 1 insertion(+)
[root@ip-172-31-41-181 mehar]#
```

- ab mn ny dykhna ha is ny kya ad kiya tha os k liy mn command ko run lro ga `git log` os k bad `git show commit ID` is sy mujy dosry bndy ka kiya hoa kam b nazr any lagy ga .

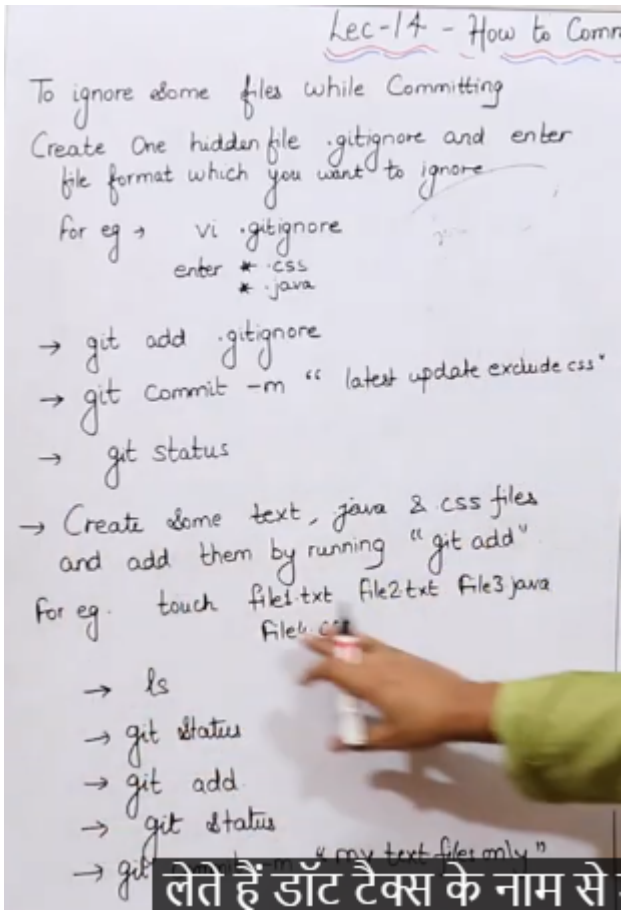
```
+jinhi malom hi nhi wafaa kya ha .
[root@ip-172-31-41-181 mehar]#
```

yha pr b ya data mnazr a rha ha k os bndy ny ya commit kiya tha .

- ab mn ny agr kisi file ko egnore krna ha to mn os k liyy ak file bnao ga jo `.gitignore` k name sy ho gi jysy myry pas 100 files hn mn chahta ho k myri 100 mn sy 70 file ko ignore krna ha to mn os `.gitignore` file k andr os ko add kr do ga . or jb mn os file ko run kro ga to wo in 70 file ko ignore kr k 30 ko chla dy ga .

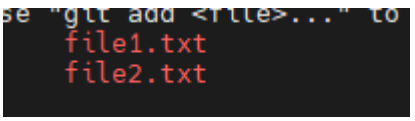
- mn file ko bnany k liyyy `vi .gitignore` ki command ko run kro ga . hmary pas jitni b `.css,.java` jysi files hn on ko ignore kr skty hn .

- is k bad mn ny `git add .gitignore` is ko add kr liya . is ko hm staging earia mn ly gy .

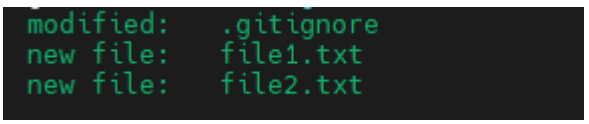


- mn ny new file bnai os k andr mn ny do file ky name ko add kiya `vi .gitignore` sy file k ander gya or os mn `* .css , * .java` ko add kiya is k bad mn ny `git add .gitignore` ki command ko run kiya .

- is k bad mn ny commit krna ha is k liyy mn `git commit -m "ignore file format"` ya command ko run kro ga .

- jb mn ny `git status` kiya to mujy .css .java wali file nazr nhi ai 

jo dosri file the whi nazr ai hn.yani in file ko ignore kr diya gya ha

- ab mn in file ko add kro ga `git add .` 

myri new file b add ho gai hn . ab mn `git commit -m "ignore file java and css"` is k bad mn ny `git log` sy dykhna ha k new commit ID ai ha k nhi . jb mn koi .java ya .css wali file bnao ga wo show nhi ho gi na hi commit ho gi git os ko ignore kr dy ga .

- agr mn ny dykhna ha k mn ny last kitny commit kiyy hn 1 ,2 ,3 is k liyy command ko run kro ga `git log 1,2,3,4,5,6` kitni counting likho ga otny hi commit nazr ayy gy jysy k mn ny dykha

```
[root@ip-172-31-41-181 mehar]# git log -4
commit 458dbeeb43296a7cdcec478be8996ec5a18984e3 (HEAD -> master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 08:49:41 2026 +0000

    ignore files java and css

commit 53e54d7ff8d63e1c8f6b9adcba0f8f337bc31d6a
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 08:34:19 2026 +0000

    ignore file format

commit b0ce9a05455a752eb4adeeda7d94c039cadd16e8 (origin/master)
Author: meharbadshah <meharbadshah686@gmail.com>
Date:   Mon Oct 19 06:54:57 2026 +0000

    my last data in pakistan

commit e36e2e57ffd4b2b13eab25fa8a33bbe31655c5d4
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 06:19:18 2026 +0000

    Update project data
[root@ip-172-31-41-181 mehar]#
```

- Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net mujy last waly 4 commit nazr ayy jo mn ny last mn jiyy thy.

- agr mn ny dykhna ha k mn ny sary commit ID ko ak sath hi dykh sko to is k liyy mn command ko run kro ga `git log --oneline`

```
[root@ip-172-31-41-181 mehar]# git log --oneline
458dbee (HEAD -> master) ignore files java and css
53e54d7 ignore file format
b0ce9a0 (origin/master) my last data in pakistan
e36e2e5 Update project data
4c96160 Update project data
7d4999d Update project data
84cb93c c7304c99beec65bc79c4c7be8a6468a8574d0816
c7304c9 Merge branch 'master' of https://github.com/MeharBadshah5217/central-repositery
09a647d my second repo commit
fceccfd Add content to akhir file
3106e90 my second repo in pakistan
3dd5a78 my first commit for badshah
[root@ip-172-31-41-181 mehar]#
```

or hmy sary commit ID nazr a jay gi .

- agr mn koi apna commit bhol gya ho or mn ny apna commit dondna ha to mn `git log --grep "ignore"` ko search kro ga mujy koch is trha ka nazr ay ga k mn ny 2 commit kiyy thy k jis mn ignore

```
[root@ip-172-31-41-181 mehar]#  git log --grep "ignore"
commit 458dbeeb43296a7cdcec478be8996ec5a18984e3 (HEAD -> master)
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 08:49:41 2026 +0000

    ignore files java and css

commit 53e54d7ff8d63e1c8f6b9adcba0f8f337bc31d6a
Author: meharumar <meharbadshah5218@gmail.com>
Date:   Mon Oct 19 08:34:19 2026 +0000

    ignore file format
[root@ip-172-31-41-181 mehar]#
```

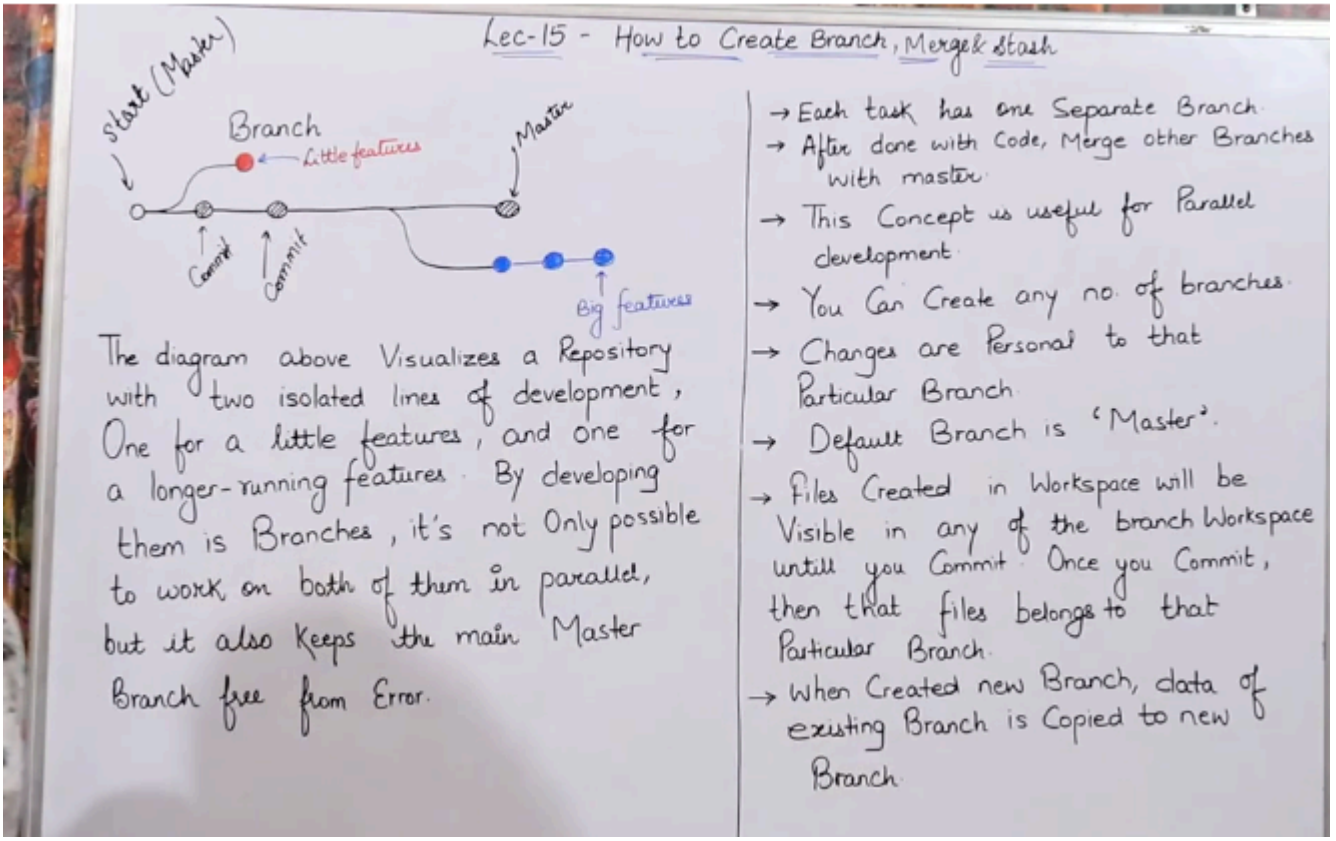
aya tha . mn jo massege ko add krna ho os sy b mn dondh skta ho .

خبر سن کر میرے مرنے کی، وہ بولے رقیبوں سے

خدا بخشے، بہت خوبیاں تھیں مرنے والے میں۔

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lecture NO: 5 - How to create Branch in git| What is git reset,git stash and git Merge | What is git conflict



Master branch ?

master branch project ki main branch hoti hai, jahan project ka stable ya integrated code rakha ja sakta hai.

- branch bnany sy hm multipal kam kr skty hn jis wajha sy hmari master branch py koi fark nhi prdy ga .

- jb new branch bnti ha to hmari jo master branch hoti ha os ko koi frk nhi prdy ga os ka pora data next branch mn coppy ho jay ga . ab ya bnda apny cord bnata jay ga 1 ya is sy zyada cord bnay ga . agr is ny cord bnaty bnaty koi garbard kr b di to hmy koi frk nhi prdy ga ku k hm ny is ko 1 alyda branch bna k di hovi ha .
- hm chahty hn k ak application ko kon sab sy acha dazin kr skta ha os k liyy hmary pas cord hona chahyy jo hmari main branch macter branch k pas hota ha . or hm 10 bndo ko apna cord dy gy or on sab ko khy gy k wo apna apna kam kry or cord ko achy sy prfom kry . jo sab sy acha prfom kry ga hm os ka cord marge kry gy master branch k sath. or jo bndy ny new task prfom kiya ha wo master branch mn marge ho jay ga.
- hm ny ab tk jo b kam kiya ha wo hmari master branch mn kam hoa ha .
- agr myry pas 2 bndy hn mn 2 branch bna lo ga child branch agr myry pas master branch mn koi cord ha wo baki 2 mn b coppy ho jay ga . mn ny agr koi new file bnai ha `xyz` to mujy wo 3 branch mn nazr ayy gi jb tk mn os ko git add nhi krta ya commit nhi krta tb tk sab k liyy same nazr ayy ga .

Dosri branch ?

Doosri branch mein hm naya feature **develop** kar sakte hain ya **bug fix** kar sakte hain, bina main branch ke code ko directly change kiye.

What is git reset ?

git reset ka use commits ko peeche le jane aur changes ko staging area ya working directory se remove karne ke liye hota hai.

What is git stash ?

Jab aap kisi project par kaam kar rahe hon aur changes commit kiye baghair temporarily side par rakhna chahte hon, to git stash use karte hain

What is git merge ?

Jab aap ek branch ka code doosri branch mein combine karna chahte hain, to git merge use karte hain.

What is git conflict ?

Jab do branches mein ek hi file ki same lines par different changes hon aur Git khud decide na kar sake ke kaunsa change rakhna hai, to merge conflict hota hai.

- agr mn ny dykhna ha k mn konsi branch ho to mn `git branch` ko command ko run kro ga mujy os ka output `*master` mily ga .
- jis branch k agy * laga hota ha is ka matlab hota ha k mn is branch mn hota ho .
- Agr mn ny new branch bnani ha to `git branch branch1` to mn ny new branch bna li ha os ka name rakha ha **branch1** .
- agr mn ny apni **master branch** sy bhr jana ha or **branch1** mn jana ha to mn ak command ko run kro ga `git chekout branch1` ab mn apni master branch sy nikl kr branch1 mn a jao ga .

- Agr mn ny apni kisi brnach ko delete krna ha to is k liyy mn command ko run kro ga `git branch -d branch name` .
- Agr mn ny koi branch delete krni ha or myra koi kam chal rha ha to mn os ko forcfully dellete krna ha to `git branch -D branch name` ka istmal kro ga .
- Agr mn ny dykhna ha mn ab kis branch mn ho to mn ya command ko run kro ga . `git branch` agr mn ny new branch bnai ha or os ko b dykhna ha to mn ya hi command ko run kro ga

```
[root@ip-172-31-46-110 mehar]# git branch
  branch1
* master
[root@ip-172-31-46-110 mehar]# git chekout branch1
```

- mn ny apni master branch sy dosri branch1 mn jana ha to mn ny `git checkout branch1` ki command ko run kro ga

```
[root@ip-172-31-46-110 mehar]# git checkout branch1
Switched to branch 'branch1'
[root@ip-172-31-46-110 mehar]# git branch
* branch1
  master
[root@ip-172-31-46-110 mehar]#
```

ab mn branch1 mn a gya ho .

- ab mn branch1 mn new file bnata ho `branch1file` os k andr koi data likho ga .

```
[root@ip-172-31-46-110 mehar]# cat >branch1file
helo to all [root@ip-172-31-46-110 mehar]#
[root@ip-172-31-46-110 mehar]# ls
akhir  badshah  branch1file
```

mn ny branch1 mn new file bna di ha jis sy myri master branch mn koi frk nhi parda . ab mn is ko git mn add kro ga `git add .` ya command ko run kro ga .

```
[root@ip-172-31-46-110 mehar]# git log --oneline
b0ce9a0 (HEAD -> branch1, origin/master, master) my last data in pakistan
e36e2e5 Update project data
```

myry pas mujy new branch mn jo add kiya ha wo nazr a rha ha .

- Agr mn ny 1 branch sy dosri branch mn jana ha to mn `git switch branch name` ko run kro ga .
- mn jb tak apny cord ko commit nhi krta tb tk mujy master branch or dosri branch mn nazr ata rhy ga .
- Agr mn ny apni kisi branch ko marge krna ha master branch k sath to mn ya cord ko run kro ga `git merge <branch name >` .
- mn ny branch 1 ko master branch k sath merge krny k liyy ya kam kiya

```
[root@ip-172-31-46-110 mehar]# git merge branch1
Updating b0ce9a0..0b25e1e
Fast-forward
 branch1     | 1 +
 branch1file | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 branch1
 create mode 100644 branch1file
[root@ip-172-31-46-110 mehar]#
```

- Agr mn apni 2 brancho mn 1 hi name sy 2 file bna lo to kam hota rhy ga lakin jb mn in ko merge kro ga to confilict ayy ga is liyy same name sy file nhi bnani.

- agr myry pas koi 2 file same hn lakin 1 file mn likha ha `myr name is umar` or dosri file mn likhaa ha `my name is umar munir` to ya ho ga k myry pehly waly data mn sirf wo wali changing ko hi add kry ga jo dosri file mn ha `munir` is ko add kr ly ga .

- conflict is time ay ga jb 1 file mn `my name is umar` or 2 mn likha ho ha `i am devops engener` tpo is ko conflict ayy ga . is ko samjh nhi ayy ga k pehly data konsa likhna ha kis ko opr likhna ha konsa nichy likhna ha .

- mn ny 1 master branch mn file bnai ha os add kiya ha pr commit kiya ha os k bad mn apni branch1 mn jao ga os mn 1 file bnao ga same name sy os k andr mn data likho ga .

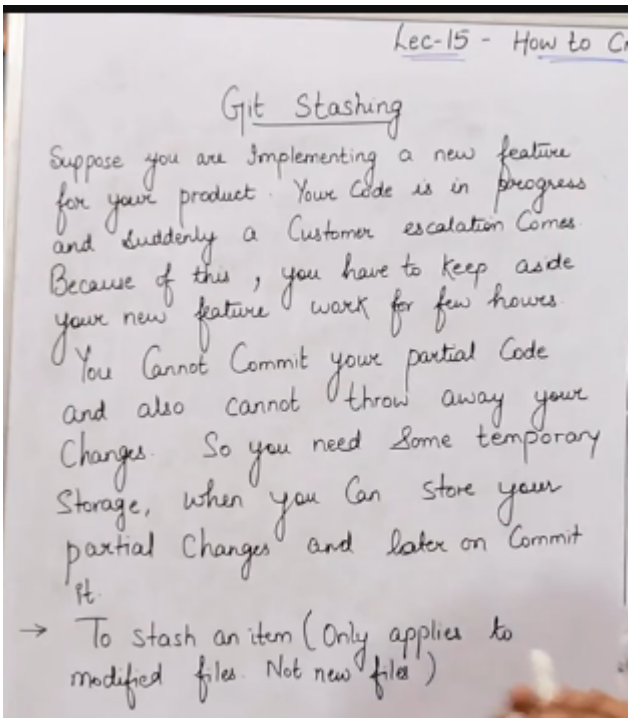
- mn ny dono side file bnai or on ka data same likha lakin kisi wajha sy conflict a gya mn master branch mn rh ke os file ko `vi meharsahab` sy edit kro ga or pr os mn koi changing kr do ga to conflict khatm ho jay ga .

- mn ab is ko add kro ga `git add .` is k bad mn `git commit -m "conflict resolve` ki command ko run kro ga .
```
[root@ip-172-31-46-110 mehar]#   git commit -m "conflict resolve"
[master 935c925] conflict resolve
[root@ip-172-31-46-110 mehar]#
```
mujy ya nazr ayy ga .

- hmy merge krny ki zarort nhi ha bs hm ny ya hi btana tha k konsi chez nichy dyni ha konsi oper dyni ha .

- is k bad mn `git log --oneline` ki command ko run kro ga .
```
[root@ip-172-31-46-110 mehar]# git log --oneline
935c925 (HEAD -> master) conflict resolve
b516699 (branch1) commit from master
108dcae my data meharsahab
0b25e1e (origin/master, origin/branch1) Add new changes to dev
b0ce9a0 my last data in pakistan
e36e2e5 Update project data
4c96160 Update project data
7d4999d Update project data
84cb93c c7304c99beec65bc79c4c7be8a6468a8574d0816
c7304c9 Merge branch 'master' of https://github.com/MeharBadshah5217/central-repositery
09a647d my second repo commit
fceccfd Add content to akhir file
3106e90 my second repo in pakistan
3dd5a78 my first commit for badshah
[root@ip-172-31-46-110 mehar]#
```

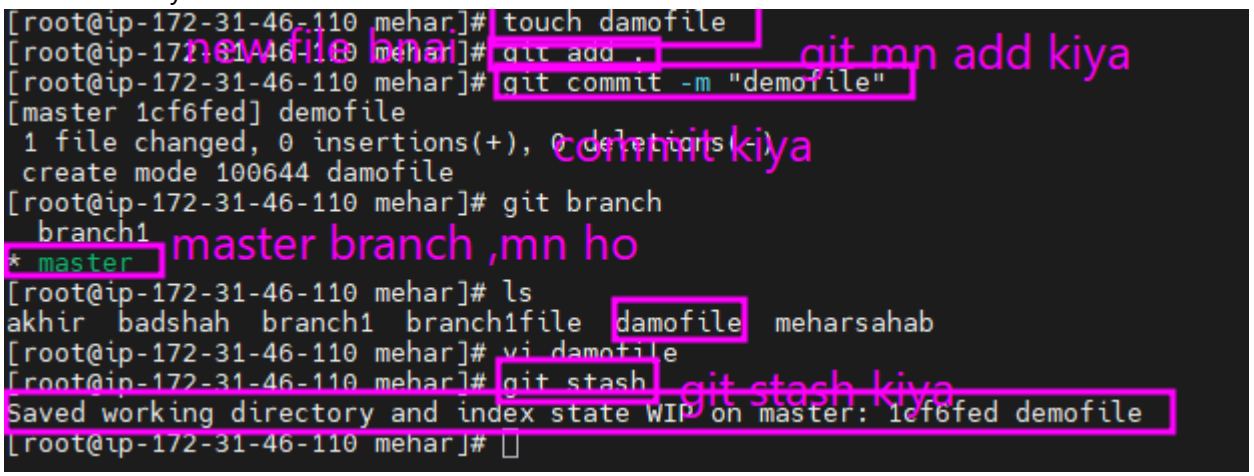
mujy ya nazr a rha ha jo mn ny changing ki ha .



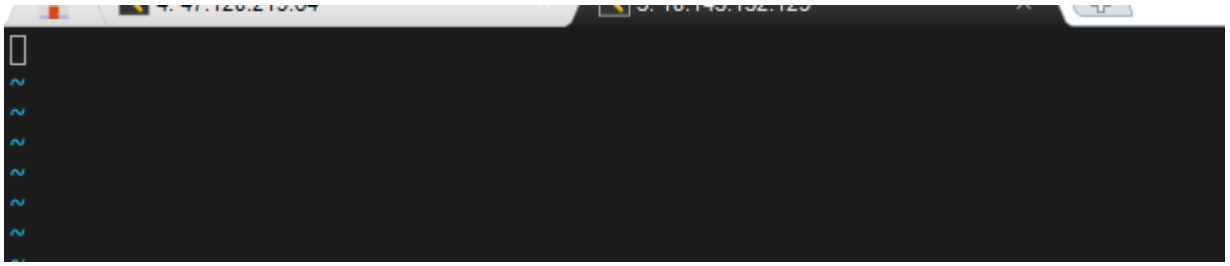
Git stashing ?

myry pas ak branch mn kam ho rha ha or os ka cord mn ny 90% likh liya ha or mujy koi zarori kam prd gya ha mn chahta ho k mn apna cord pora likh kr hi commit kro ga to mn is k liyy `git stash` kro ga is sy hmary pas data stor ho jay ga . jisi or jga pr or mn apnoi local repo mn koi jo new kam aya ha os ko kr lo ga kam pora krny k bad mn apna jo cord stash kiya tha os ko waps kr lo ga is k liyy commands ko use krty hn .

- jb mn `git stash` ki comand ko run kro ga to myri local repo khali ho jay gi .
- mn dykhna chahta ho k myri git stash mn kon konsi files pardi hn on ko dykh lo `git stash list` ki comand ko run kro ga .
- mn ny 1 file bnai ha os ka name rakha ha `demofile` os k bad mn ny is kpo add kiya pr commit kiya . pr is k bad mn ny is file k ander koi data likha ha



is mn jo nazr a rha ha ,mn ny ab git stash kr diya ha to git stash sy myri local repo mn koi chez nhi ha ab wha pr sari jga free ho gai ha . ab mn dobara sy apni demofile ko khalo ga to os mn koi chaez nazr nhi ay gi .


is file ka data stash mn chala gya ha . mn ny dykha ha k kitni file stash mn hn . is k liyy `git stash list` ki comand ko run kro ga .

```
[root@ip-172-31-46-110 mehar]# git stash list
stash@{0}: WIP on master: 1cf6fed demofile
[root@ip-172-31-46-110 mehar]#
```

mujy ya nazr ayy ga . is k bad mn ny apni file k andr likhna ha second cord from clinet .

```
[root@ip-172-31-46-110 mehar]# git stash
Saved working directory and index state WIP on master: 1cf6fed demofile
[root@ip-172-31-46-110 mehar]# git stash list
stash@{0}: WIP on master: 1cf6fed demofile
stash@{1}: WIP on master: 1cf6fed demofile
[root@ip-172-31-46-110 mehar]#
```

mn ny pr sy git stash kiya to myry pas pr sy file k andr ka data khali ho gya or file k andr koch b nhi ha .

- ab mn ny apna prana data ko dobara sy apni fiole mn ly k ana ha to mn ya commad ko run kro ga `git stash apply stash@{0}` to myra prana data waps a gyta ha .

Git reset kya ha ?

mn ny agr koi kam apny working aria mn kiya tha or os ko mn ny add kr diya mujy bad mn yad aya k mn ny abi is mn koch or kam krna tha to os ko dobara sy staging aria mn sy remove krna ha to is jga pr git reset ko use kro ga .is k liyy mn `git reset file name` is k bad hm `git reset .` ki comad ko run kry gy .

```
[root@ip-172-31-46-110 mehar]#  cat >file6.txt
hello bajoi ki hal a
[root@ip-172-31-46-110 mehar]# git add .
[root@ip-172-31-46-110 mehar]# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   damofile
        new file:   file6.txt

[root@ip-172-31-46-110 mehar]# git reset .
Unstaged changes after reset:
M       damofile
```

is mn ny new file bnai ha os k bad mn ny os ko add kiya ha . or myri new changing green ho gai yani add ho gai ha . lakin mujy yad aya ha k mn ny is ko abi staging aria mn nhi dalna tha . to mn ab icxs k liyy `git reset .` ya command ko run kiya to mujy pr sy status mn jo changing ki the wo red nazr any lagy gi

```
[root@ip-172-31-46-110 mehar]# git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   damofile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file6.txt

no changes added to commit (use "git add" and/or "git commit -a")
[root@ip-172-31-46-110 mehar]#
```

mn agr

is ko pr sy `git add .` kro ga to dobara sy git mn add ho jay ga .

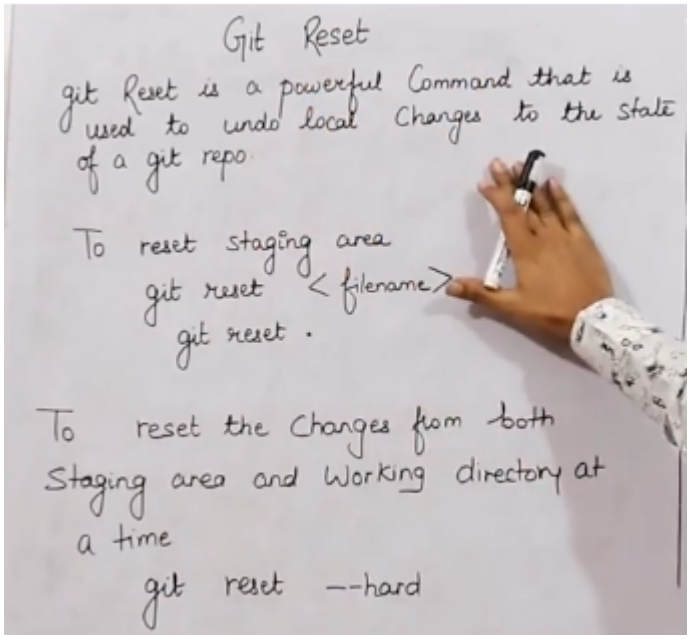
```
[root@ip-172-31-46-110 mehar]# git add .
[root@ip-172-31-46-110 mehar]# git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   damofile
        new file:   file6.txt

[root@ip-172-31-46-110 mehar]#
```

- agr mn ny apni working space sy or staging aria sy cord ko remove krna ha to mn `git reset --hard` ya comand ko run kro ga .

```
[root@ip-172-31-46-110 mehar]# git reset --hard
HEAD is now at 1cf6fed demofile
[root@ip-172-31-46-110 mehar]#
```

mn ny pr sy `git status` kiya to mujy koch b nazr nhi ayy ga.

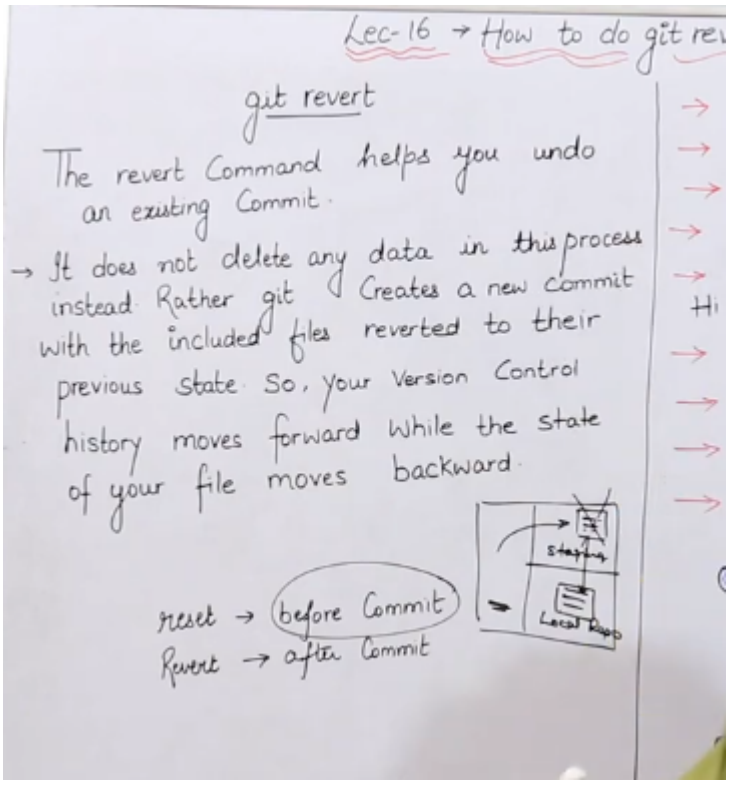


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lecture No : 6 = What is git revert |What is tag in git | Github clone | git reset vs git revert | gitlab

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Git revert kya ha ?




```
git revert aik aisa command hai jo purane commit ki changes ko khatam karne ke liye aik naya commit banata hai. Yeh pichli history ko delete nahi karta, is liye team projects ke liye safe ha.
```

- Agr hmary pa koi cord tha hm ny os ko `git add .` kr liya is k bd hm ny os cord ko `git commit -m "project AS` commit b kr diya ha to is mamly mn hm git revert ko us krty hn or jo hm ny commit kiya ha os ko git revrt kr dy gy is sy ya ho ga k hmara commit kiya hoa cord a,b,c,d ha or myra pehly wala data a,b,c tha to jb ya command ko run kiya to hmara cord jo commit kiya tha oki jga pr jo hmari mater branch ka cord tha wo commit ho jay ga os ki jga pr hmara prana data a jay ga .

Untracked file ?

untracked files woh files hoti hain jo abhi Git ke tracking system mein add nahi hui hotin.

- `git clean -n` is command sy hm jo unuse file hn on ko delet kety hn laikin ya hn s pochy ga k delte kna h ak nhi .
- Agr `git clean -f` ya command sy force fully delete ho jay gi file .

Tags

hamry pas 10 commit ID hn or hm ko nhi pata k konsi commit ID mn kya cord likha ha or agr mn ny os ki pechan krni ha to mn os ky opr tage lga do ga jb b hm commit krty hn to sath tage b lgaty hn jysy `git commit -m "my docker cord"` is commit mn mn ny **"my docker cord"** ya tage diya ha k is commit ID mn myra docker ka cord prda ha .

- hmary pa 10 commit ID b ho to hm sab ko alyda alyda tags dy kr sab ki pehchan kr skty hn .
- Agr mn ny apny sary tags dykhny hn to mn `git tag` ya command sy sary tage ko dakh skta ho.

- Agr mn ny content ko dykhna chahta ho tags ki madad sy to mn `git show tage name` sy mn commit Id ko b dykh skta ho .
- Agr mn ny apna tags ko delete krna ha to mn `git tag -d tag nam` sy delte k skta ho apna tage .

Github clone

Git clone ek command hai jo remote repository (jaise GitHub) ki copy apne local computer ya server par download karne ke liye use hoti hai. Is mein project ki files, branches aur commit history bhi copy hoti hai.

- `git clone https://github.com/username/repository.git` is trha hn git clone krty hn .

Git branch , Github branch mn kya frk ha ?

- git mn hm branch bnaty hn or master branch ka data chlid branch mn b coppy ho jata ha .ya sara kam hn command sy krty hn .
- github mn b hm branch bnaty hn or master branch ka sara data child branch mn coppy ho jata ha lakin is mn ya fark ha k hm 1 hi click mn sara kam kr lyty hn . ku k github **GUI** ha or git **CLI** ha .
- jysy git mn master branch k sath child branch ko merg krty hn vysy hi hn github mn master branch k sath child branch ko merg kr skty hn .

Pulling rquest ?

Agr hmary pas to 2 branches hn or hm ny on ko mecrg krna ha .hmy nhi pata k jo hm ny file bnai ha os mn koi conflict ha k nhi to hm os sy pehly pulling request byj k conferm krty hn k koi conflict to nhi ha .

- mn ny apni ec2-instance ko run kiya or apni working dericatory mn gya or wha pr ak file bnai `cat >rvetfile` k nam sy os k and mn ny likha `paneer` . or mn ny add kiya `git add .` or pr commit kr diya `git commit -m "this is first shahi prata"` .

```
[root@ip-172-31-46-110 mehar]# git commit -m "this is mn my first alo pratha
> ^C
[root@ip-172-31-46-110 mehar]# git commit -m "this is mn my first alo pratha"
[master 9d29190] this is mn my first alo pratha
 1 file changed, 1 insertion(+)
 create mode 100644 revertfile
[root@ip-172-31-46-110 mehar]#
```

- mn ny pr sy apni `rvertfile` mn likha `pyaz`

```
[root@ip-172-31-46-110 mehar]# git log
commit 2bb0e2a4853069cd53f0cbfcc0cd41ab1873d26d (HEAD -> master)
Author: MeharUmarmunir <meharsahab5218.com>
Date:   Tue Oct 20 06:49:17 2026 +0000

    this is pyaz

commit 9d29190c531ab6d3fae69a4fcfcd9292149499c6
Author: MeharUmarmunir <meharsahab5218.com>
Date:   Tue Oct 20 06:45:12 2026 +0000

    this is mn my first alo pratha
```

mn ny 2 commit kiyy hn

```
[root@ip-172-31-46-110 mehar]# cat revertfile
pyaz    pehly mn ny pyaz likha tha jo sai tha .

alo     ,ab mn ny alo likha ha jo k sai nhi ha .
[root@ip-172-31-46-110 mehar]#
```

jb mn ny `git revert`

`commit ID` likhi to mujy koch aysa nazr aya

```
Revert "this is alo"

This reverts commit 7fc06aa8747aac0c2236e34840403fc61cf3e664.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# Changes to be committed:
#       modified:   revertfile
#
```

- jb mn ny apni dosri comit ki the os mn ali likha ha to mn ny abb os ki jga pr pyaz ko likh diya ha.

- Agr agr myry pas 10 files hn or mn on ko remove krna chahta ho on ka ko kam nhi ha . `touch file.a file.b file.c file.d file.e file.f file.i` sy file bnai hn .

```
[root@ip-172-31-46-110 mehar]#  git  status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file.a
        file.b
        file.c
        file.d
        file.e
        file.f
        file.i

nothing added to commit but untracked files present (use "git add" to track)
[root@ip-172-31-46-110 mehar]#
```

ya untrack file hn .

agr mn ya command ko run kro `git clean -n` to mujy koch aysa nazr ayy ga

```
[root@ip-172-31-46-110 mehar]# git clean -n
Would remove file.a
Would remove file.b
Would remove file.c
Would remove file.d
Would remove file.e
Would remove file.f
Would remove file.i
[root@ip-172-31-46-110 mehar]#
```

poch rha ha remove krna ha

is k bad mn agr `git clean -f` ki command ko run krta ho to mujy ya koch aysa nazr ay ga .

```
[root@ip-172-31-46-110 mehar]# git clean -f
Removing file.a
Removing file.b
Removing file.c
Removing file.d
Removing file.e
Removing file.f
Removing file.i
[root@ip-172-31-46-110 mehar]#
```

is ka matlab ha file delete ho gai .

.

git tags kysy krty hn ..

- `git log --oneline` is command sy hmy sari commit ID nazr ayy gi .

```
[root@ip-172-31-46-110 mehar]# git tag -a inportant -m "this is very iportent tag" 7fc06aa8747aac0c2236e34840403fc61cf3e664
```

mn ny ya command ko run kiya os ky bad mn ny

```
[root@ip-172-31-46-110 mehar]# git log --oneline
01f2794 (HEAD -> master) Revert "this is alo" "peleas revert my ID"
7fc06aa (tag: inportant) this is alo
2bb0e2a this is pyaz
9d29190 this is mn my first alo pratha
1cf6fed demofile
935c925 conflict resolve
b516699 (branch1) commit from master
108dcae my data meharsahab
0b25e1e (origin/master, origin/branch1) Add new changes to dev
b0ce9a0 my last data in pakistan
```

ab mujy is commit k sath tage laga nazr a rha ha .

- github sy hm kysy apni local mn clone krty hn is k liyy hm command ko run kry gy `git clone https://github.com/MeharBadshah5218/central-repositery.git` or hmary pas github sy sara data hmar pas local mn clone ho jay ga . or ab mn `ls` kr ky dykho to mujy koch is trha naz y ga .

```
[root@ip-172-31-46-110 ec2-user]# ls
central-repositery  mehar
[root@ip-172-31-46-110 ec2-user]#
```

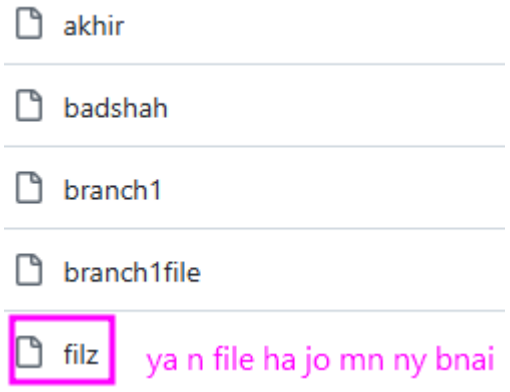
ya myry pas **cnteral-repository** github sy clone ho gai ha .

- agr mn npni `cnteral-repository` ky and jao to mujy is k andr sara data nazr ay ga.

```
[root@ip-172-31-46-110 ec2-user]#  cd central-repositery/
[root@ip-172-31-46-110 central-repositery]# ls
akhir  badshah  branch1  branch1file
```

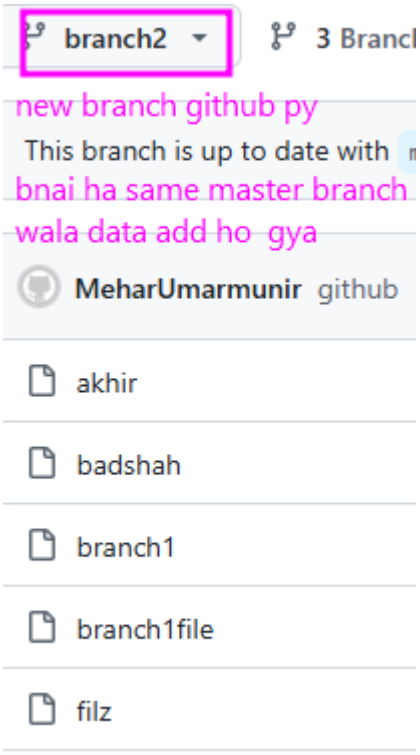
cnteral-repository

- mn ny 1 new **file** bnai or o mn ny **add** kiya pr **commit** kiya pr **push** kiya .

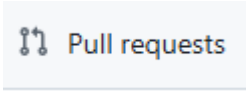


- mn ny apni github mn dykha to mujy ha pr push ki hovi file nazr a gai ha .

- is k bad mn ny apny github k andr hi 1 branch bnai to myri master branch ka sara data os new branch



mn coppy ho gya .

- agr hm ny koi cord likha ha or is ko mn ny direct push nhi krna blky pehly pull requst byj k chek krna ha to  hmy github py koch is trha ka nazr ayy ga .